



Qualification Pack

Advanced Course in Computer Application

Advanced Programming: Programming with Python/ Advanced Programming: Core Java

QP Code: HAR/ITS/Q1602

Version: 1.0

NSQF Level: 5

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Qualification Pack

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HAR/ITS/Q1602: Advanced Course in Computer Application

Brief Job Description

Advanced Course in Computer Application course, designed to impart comprehensive knowledge and practical skills covering all aspects of computer uses in business, industries and service sectors.

Personal Attributes

Strong analytical thinking, problem-solving ability, attention to detail, creativity, teamwork, communication skills, adaptability, and willingness to learn new technologies.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HAR/ITS/N0909: Basics of Information Technology](#)
2. [HAR/ITS/N0407: Operating System \(ACCA\)](#)
3. [HAR/ITS/N1604: Introduction to Data representation and Digital Logic](#)
4. [HAR/ITS/N1204: Basics of Web Page Designing \(45 hrs.\)](#)
5. [HAR/ITS/N1603: Data Communications and Networking Concepts](#)
6. [HAR/ITS/N1601: Object-Oriented Programming and Database Management System](#)
7. [HAR/ITS/N1602: Introduction to Software Development Life Cycle \(SDLC\)](#)
8. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
9. [HAR/ITS/N1610: OJT/Project & Viva \(60 hrs.\)](#)

Electives (mandatory to select at least one):

Elective 1: Advanced Programming: Programming with Python

1. [HAR/ITS/N1608: Programming with Python \(ACCA\)](#)

Elective 2: Advanced Programming: Core Java

1. [HAR/ITS/N1609: Core Java](#)



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Qualification Pack (QP) Parameters

Sector	Information Technology Sector
Sub-Sector	IT- Services
Occupation	Software Developer
Country	India
NSQF Level	5
Credits	21
Aligned to NCO/ISCO/ISIC Code	2512.0204
Minimum Educational Qualification & Experience	Pursuing 3rd year of UG OR Completed 2nd year of UG (UG Diploma)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	18/02/2028
NSQC Approval Date	18/02/2025
Version	1.0
Reference code on NQR	QG-05-IT-04415-2025-V2-HART
NQR Version	2

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HAR/ITS/N0909: Basics of Information Technology

Description

The NOS provides basic knowledge and skills to operate computer and prepares you to perform basic office jobs.

Scope

The scope covers the following :

- The NOS builds essential skills in Word, Excel, PowerPoint, and Internet usage, preparing students for basic office tasks and serving as a foundation for advanced computer applications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the meaning, characteristics, history, generations, and classification of computers.
- PC2.** Identify and explain the main components, input/output devices, and the working of a computer system.
- PC3.** Differentiate between data and information
- PC4.** Create and edit documents using word processing tools with proper formatting and layout.
- PC5.** Apply formulas and functions in spreadsheets and use spreadsheet tools effectively.
- PC6.** Design and present information visually using digital presentation software.
- PC7.** Utilize networking and internet skills to communicate online, manage email accounts, ensure cybersecurity, and understand cloud storage

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Demonstrates understanding of the evolution, characteristics, and classification of computers, including their historical development and technological generations.
- KU2.** Identifies and explains the main hardware components, input/output devices, and the overall functional process of a computer system.
- KU3.** Demonstrates knowledge of digital productivity tools, including word processors, spreadsheets, and presentation software, for creating and managing professional documents and visual materials.
- KU4.** networking concepts, internet services, communication protocols, and the applications of the internet and IECT.
- KU5.** using web browsers, search engines, Wi-Fi, Bluetooth, and online services for browsing, communication, uploading, downloading, and accessing e-governance platforms.
- KU6.** to apply internet safety measures including strong passwords, antivirus tools, firewalls, safe browsing practices, and awareness of cyber threats and online privacy issues.



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KU7. cloud computing concepts, types of cloud storage, benefits of online storage, and usage of cloud storage service providers for data management.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. set goals for career development.

GS2. efficiently plan daily data entry tasks to meet deadlines.

GS3. Ensure accuracy and confidentiality of customer data.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	11	15	10	-
PC1. Understand the meaning, characteristics, history, generations, and classification of computers.	-	-	-	-
PC2. Identify and explain the main components, input/output devices, and the working of a computer system.	-	-	-	-
PC3. Differentiate between data and information	-	-	-	-
PC4. Create and edit documents using word processing tools with proper formatting and layout.	-	-	-	-
PC5. Apply formulas and functions in spreadsheets and use spreadsheet tools effectively.	-	-	-	-
PC6. Design and present information visually using digital presentation software.	-	-	-	-
PC7. Utilize networking and internet skills to communicate online, manage email accounts, ensure cybersecurity, and understand cloud storage	-	-	-	-
NOS Total	11	15	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0909
NOS Name	Basics of Information Technology
Sector	Information Technology Sector
Sub-Sector	
Occupation	Data Entry Operator
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

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HAR/ITS/N0407: Operating System (ACCA)

Description

Provides a foundational understanding of operating systems, exploring their types, core functions, and architecture.

Scope

The scope covers the following :

- Equips learners with the skills necessary for roles in system administration, IT support etc.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the concepts, history, architecture, functions, and types of operating systems including Windows, Linux, Android, iOS, and other GUI-based systems.
- PC2.** Differentiate between various operating systems such as batch, multitasking, multiprocessing, real-time, distributed, network, and mobile operating systems.
- PC3.** Describe the role of kernel, process management, CPU scheduling, deadlock handling, memory management, storage management, file systems, and security in operating systems.
- PC4.** Identify and use the elements of GUI operating systems including desktop, taskbar, start menu, icons, windows, menus, toolbars, and dialog boxes.
- PC5.** Perform file and folder management operations in Windows operating system including creating, copying, moving, renaming, and deleting files and folders.
- PC6.** Use Control Panel settings and system utilities for managing basic system configurations and devices.
- PC7.** Connect and configure peripheral devices such as keyboard, mouse, monitor, printer, projector, and scanner with a computer system.
- PC8.** Apply basic troubleshooting techniques to resolve common hardware and software issues related to power supply, printer, scanner, and peripheral devices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The objective and goals of the organization and scope of work
- KU2.** The principles of operating systems, including types of OS, process management, multithreading, and memory management.
- KU3.** How operating systems interact with hardware and manage system resources.
- KU4.** The importance of operating systems in maintaining organizational IT infrastructure, ensuring performance, security, and resource management.

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** Make informed decisions regarding operating system selection, configuration, and updates based on organizational needs and technical requirements.
- GS2.** Apply problem-solving techniques to troubleshoot operating system issues, optimize performance, and configure system resources efficiently.
- GS3.** To manage tasks effectively, ensuring timely installation, configuration, and troubleshooting of operating systems.
- GS4.** Deliver efficient and reliable operating system solutions that meet user needs and enhance organizational operation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	9	-	5	-
PC1. Explain the concepts, history, architecture, functions, and types of operating systems including Windows, Linux, Android, iOS, and other GUI-based systems.	-	-	-	-
PC2. Differentiate between various operating systems such as batch, multitasking, multiprocessing, real-time, distributed, network, and mobile operating systems.	-	-	-	-
PC3. Describe the role of kernel, process management, CPU scheduling, deadlock handling, memory management, storage management, file systems, and security in operating systems.	-	-	-	-
PC4. Identify and use the elements of GUI operating systems including desktop, taskbar, start menu, icons, windows, menus, toolbars, and dialog boxes.	-	-	-	-
PC5. Perform file and folder management operations in Windows operating system including creating, copying, moving, renaming, and deleting files and folders.	-	-	-	-
PC6. Use Control Panel settings and system utilities for managing basic system configurations and devices.	-	-	-	-
PC7. Connect and configure peripheral devices such as keyboard, mouse, monitor, printer, projector, and scanner with a computer system.	-	-	-	-
PC8. Apply basic troubleshooting techniques to resolve common hardware and software issues related to power supply, printer, scanner, and peripheral devices.	-	-	-	-
NOS Total	9	-	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N0407
NOS Name	Operating System (ACCA)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Computer Hardware and Networking Associate
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HAR/ITS/N1604: Introduction to Data representation and Digital Logic

Description

This NOS provides fundamental knowledge of data representation, binary systems, Boolean algebra, and digital logic concepts used in computer systems and digital electronics.

Scope

The scope covers the following :

- The scope includes understanding number system conversions, binary arithmetic operations, coding schemes, and error detection techniques for data representation. It also covers Boolean algebra, logic simplification methods, truth tables, logic gates, combinational logic, and implementation of digital circuits using different gate configurations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the concepts of data representation, number systems, binary arithmetic, and coding schemes used in digital systems.
- PC2.** Perform conversions between different number systems and apply binary arithmetic operations accurately.
- PC3.** Identify and explain the functions of basic and universal logic gates including AND, OR, NOT, NAND, NOR, XOR, and XNOR gates.
- PC4.** Implement digital circuits using different gate combinations and logic gate configurations such as NAND, NOR, AND-OR-INVERT, and OR-AND-INVERT.
- PC5.** Analyze and design simple combinational logic circuits using Boolean logic and logic gates.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the concepts of number systems, binary arithmetic, coding techniques, Boolean algebra, logic simplification methods, and digital logic gates used in computer and digital systems.
- KU2.** the principles of data representation, error detection and correction, combinational logic, and implementation of digital circuits using various logic gate configurations.
- KU3.** the importance of digital logic and data representation concepts in computer architecture, electronics, communication systems, and information technology applications.
- KU4.** to apply logical analysis, circuit design principles, and standard digital system practices followed in technical and computing environments.

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** Demonstrate effective oral and written communication skills for explaining concepts of data representation, Boolean algebra, and digital logic circuits.
- GS2.** Apply analytical thinking and problem-solving skills while performing number system conversions, logic simplification, and circuit design tasks.
- GS3.** Exhibit professionalism, attention to detail, teamwork, and logical reasoning while working on digital logic and data representation activities.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	9	-	5	-
PC1. Explain the concepts of data representation, number systems, binary arithmetic, and coding schemes used in digital systems.	-	-	-	-
PC2. Perform conversions between different number systems and apply binary arithmetic operations accurately.	-	-	-	-
PC3. Identify and explain the functions of basic and universal logic gates including AND, OR, NOT, NAND, NOR, XOR, and XNOR gates.	-	-	-	-
PC4. Implement digital circuits using different gate combinations and logic gate configurations such as NAND, NOR, AND-OR-INVERT, and OR-AND-INVERT.	-	-	-	-
PC5. Analyze and design simple combinational logic circuits using Boolean logic and logic gates.	-	-	-	-
NOS Total	9	-	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1604
NOS Name	Introduction to Data representation and Digital Logic
Sector	Information Technology Sector
Sub-Sector	
Occupation	Software Developer
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HAR/ITS/N1204: Basics of Web Page Designing (45 hrs.)

Description

Provides knowledge and skills to create web pages using HTML, CSS and JavaScript

Scope

The scope covers the following :

- The NOS covers the fundamental skills required to create, design, and validate interactive web pages.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Create web pages using HTML and CSS.

PC2. Use HTML elements, tags & attributes.

PC3. Create lists, links, and tables using HTML tags

PC4. Make WebPages more attractive using CSS

PC5. Understand JavaScript basics- variables, functions, and control structures

PC6. Create interactive web pages using Javascript.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Basic structure of web pages (HTML elements, tags, attributes)

KU2. CSS styling techniques: inline, internal, and external stylesheets

KU3. Core JavaScript concepts: variables, data types, operators, loops, and functions

KU4. Debugging and error handling in JavaScript.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Analyze client requirements and design solutions that meet functional needs.

GS2. Clearly communicate ideas and requirements to team members and clients.

GS3. Manage time and priorities to meet project deadlines.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	18	15	10	-
PC1. Create web pages using HTML and CSS.	-	-	-	-
PC2. Use HTML elements, tags & attributes.	-	-	-	-
PC3. Create lists, links, and tables using HTML tags	-	-	-	-
PC4. Make WebPages more attractive using CSS	-	-	-	-
PC5. Understand JavaScript basics-variables, functions, and control structures	-	-	-	-
PC6. Create interactive web pages using Javascript.	-	-	-	-
NOS Total	18	15	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1204
NOS Name	Basics of Web Page Designing (45 hrs.)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Web Designing Assistant
NSQF Level	3
Credits	1.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

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HAR/ITS/N1603: Data Communications and Networking Concepts

Description

This NOS provides knowledge of data communication systems and computer networking concepts including transmission methods, network architectures, communication protocols, reference models, and network security.

Scope

The scope covers the following :

- The scope includes understanding data communication components, transmission media, communication modes, and digital transmission methods.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the concepts, components, and applications of data communication and computer networks.
- PC2.** Identify and classify guided and unguided transmission media used in communication systems.
- PC3.** Describe the architecture, components, and types of computer networks including LAN, MAN, WAN, wireless networks, and internet networks.
- PC4.** Explain the functions and working of networking devices such as NIC, hub, switch, router, modem, bridge, repeater, and server.
- PC5.** Interpret the structure and functions of OSI and TCP/IP reference models and their respective layers.
- PC6.** Explain networking concepts related to switching, routing, multiplexing, addressing, error detection, and communication protocols.
- PC7.** Identify and describe transport, application, network access, and internet layer protocols including TCP, UDP, HTTP, FTP, DNS, SMTP, and Ethernet.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the concepts of data communication, transmission methods, network architectures, networking devices, protocols, OSI and TCP/IP models, and communication technologies used in computer networks.
- KU2.** the principles of data transmission, routing, switching, error detection, network protocols, and network security concepts including encryption, privacy, and digital signatures.
- KU3.** the role of computer networks and communication systems in organizational connectivity, information sharing, and digital communication services.
- KU4.** to apply standard networking practices, communication protocols, security measures, and network management procedures followed in IT and networking environments.



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate effective oral and written communication skills for explaining networking concepts, communication protocols, and technical issues clearly.
- GS2.** Plan and organize networking tasks, communication resources, and troubleshooting activities systematically while following technical standards.
- GS3.** Exhibit professionalism, teamwork, adaptability, attention to detail, and responsibility while working in networking and communication environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	9	-	5	-
PC1. Explain the concepts, components, and applications of data communication and computer networks.	-	-	-	-
PC2. Identify and classify guided and unguided transmission media used in communication systems.	-	-	-	-
PC3. Describe the architecture, components, and types of computer networks including LAN, MAN, WAN, wireless networks, and internet networks.	-	-	-	-
PC4. Explain the functions and working of networking devices such as NIC, hub, switch, router, modem, bridge, repeater, and server.	-	-	-	-
PC5. Interpret the structure and functions of OSI and TCP/IP reference models and their respective layers.	-	-	-	-
PC6. Explain networking concepts related to switching, routing, multiplexing, addressing, error detection, and communication protocols.	-	-	-	-
PC7. Identify and describe transport, application, network access, and internet layer protocols including TCP, UDP, HTTP, FTP, DNS, SMTP, and Ethernet.	-	-	-	-
NOS Total	9	-	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1603
NOS Name	Data Communications and Networking Concepts
Sector	Information Technology Sector
Sub-Sector	
Occupation	Software Developer
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HAR/ITS/N1601: Object-Oriented Programming and Database Management System

Description

This NOS provides knowledge and practical skills in object-oriented programming using C++, data structures, and database management systems with SQL.

Scope

The scope covers the following :

- The scope includes developing programs using C++ concepts such as functions, arrays, pointers, classes, inheritance, polymorphism, abstraction, file handling, and data structures like linked lists, stacks, queues, and trees. It also covers database concepts, ER modeling, normalization, relational database management, SQL commands, joins, triggers, views, stored procedures, and data manipulation techniques for effective database management.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the fundamentals of databases, DBMS architecture, relational models, and normalization concepts.
- PC2.** Create, manage, and manipulate databases using SQL commands such as DDL, DML, TCL, DCL, and DQL.
- PC3.** Apply SQL concepts including joins, keys, views, subqueries, triggers, and stored procedures for database operations.
- PC4.** Understand programming fundamentals, algorithms, flowcharts, and different programming paradigms.
- PC5.** Write, compile, debug, and execute programs in CPP language using control statements, functions, arrays, strings, pointers, structures, and file handling.
- PC6.** Develop applications in C++ using constructors, destructors, exception handling, dynamic memory allocation, and file handling techniques.
- PC7.** Implement data structures such as linked lists, stacks, queues, trees, and searching and sorting techniques for efficient data management.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The concepts of data, database, DBMS architecture, database models, ER models, and normalization techniques.
- KU2.** SQL syntax, commands, joins, views, subqueries, triggers, and stored procedures used in database management.
- KU3.** programming fundamentals including algorithms, flowcharts, programming paradigms, and translators.

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- KU4.** The concepts of structured and object-oriented programming including classes, objects, inheritance, polymorphism, abstraction, and encapsulation.
- KU5.** The role of programming and database management in software development and organizational operations.
- KU6.** How to work effectively with databases, programming tools, and development environments commonly used in the IT industry.
- KU7.** How to demonstrate awareness of professional ethics, teamwork, problem-solving, and quality practices in software development tasks.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate effective oral and written communication skills for explaining programming logic, database concepts, and technical documentation.
- GS2.** Apply analytical thinking and problem-solving skills to design algorithms, debug programs, and resolve database or application errors.
- GS3.** Plan and organize programming tasks, database activities, and project work efficiently within defined timelines and quality standards.
- GS4.** Exhibit professional behavior, teamwork, customer-centric approach, and adaptability while working in software development and IT-enabled environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	22	15	10	-
PC1. Explain the fundamentals of databases, DBMS architecture, relational models, and normalization concepts.	-	-	-	-
PC2. Create, manage, and manipulate databases using SQL commands such as DDL, DML, TCL, DCL, and DQL.	-	-	-	-
PC3. Apply SQL concepts including joins, keys, views, subqueries, triggers, and stored procedures for database operations.	-	-	-	-
PC4. Understand programming fundamentals, algorithms, flowcharts, and different programming paradigms.	-	-	-	-
PC5. Write, compile, debug, and execute programs in CPP language using control statements, functions, arrays, strings, pointers, structures, and file handling.	-	-	-	-
PC6. Develop applications in C++ using constructors, destructors, exception handling, dynamic memory allocation, and file handling techniques.	-	-	-	-
PC7. Implement data structures such as linked lists, stacks, queues, trees, and searching and sorting techniques for efficient data management.	-	-	-	-
NOS Total	22	15	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1601
NOS Name	Object-Oriented Programming and Database Management System
Sector	Information Technology Sector
Sub-Sector	
Occupation	Software Developer
NSQF Level	5
Credits	2.5
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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HAR/ITS/N1602: Introduction to Software Development Life Cycle (SDLC)

Description

This NOS provides knowledge of the Software Development Life Cycle (SDLC), its phases, development models and approach for planning, designing, developing, testing, deploying, and maintaining software applications efficiently and securely.

Scope

The scope covers the following :

- The scope includes understanding the importance, phases, and models of SDLC.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the concept, importance, benefits, and objectives of the Software Development Life Cycle (SDLC).
- PC2.** Identify and describe the phases of SDLC including planning, requirement analysis, design, coding, testing, deployment, and maintenance.
- PC3.** Differentiate between various SDLC models
- PC4.** Prepare basic software development documentation related to requirements, design, testing, and maintenance activities.
- PC5.** Analyze software development workflows and recommend suitable SDLC practices for efficient project management.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The concepts, phases, models, methodologies, and best practices of the Software Development Life Cycle (SDLC)
- KU2.** software planning, requirement analysis, system design, coding, testing, deployment, maintenance processes, and security considerations in software development projects.
- KU3.** the role of SDLC in managing software development projects, ensuring software quality, and improving organizational efficiency.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate effective oral and written communication skills for discussing software requirements, development processes, and project documentation.
- GS2.** Apply analytical thinking and problem-solving skills while planning, designing, testing, and maintaining software applications.



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- GS3.** Plan and organize software development activities, documentation, testing schedules, and project tasks efficiently while meeting quality standards and timelines.
- GS4.** Exhibit professionalism, teamwork, adaptability, attention to detail, and responsibility while working in software development and project environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	9	-	5	-
PC1. Explain the concept, importance, benefits, and objectives of the Software Development Life Cycle (SDLC).	-	-	-	-
PC2. Identify and describe the phases of SDLC including planning, requirement analysis, design, coding, testing, deployment, and maintenance.	-	-	-	-
PC3. Differentiate between various SDLC models	-	-	-	-
PC4. Prepare basic software development documentation related to requirements, design, testing, and maintenance activities.	-	-	-	-
PC5. Analyze software development workflows and recommend suitable SDLC practices for efficient project management.	-	-	-	-
NOS Total	9	-	5	-

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National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1602
NOS Name	Introduction to Software Development Life Cycle (SDLC)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Software Developer
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025



Qualification Pack

HAR/ITS/N1610: OJT/Project & Viva (60 hrs.)

Description

The NOS enabling students to apply their acquired skills and knowledge in real work or job-oriented environments.

Scope

The scope covers the following :

- Provides practical exposure, enhance skills, and prepare students for real work environments by bridging the gap between theory and practice

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrates application of theoretical knowledge in practical tasks

PC2. Adapts to workplace requirements and follows instructions

PC3. Demonstrates clear understanding of concepts during viva

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Workplace practices and professional standards

KU2. Basic problem-solving techniques

KU3. Communication skills in a work environment

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Work collaboratively as part of a team

GS2. Follow instructions and organizational procedures

GS3. Manage time and meet deadline

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	-	30	30
PC1. Demonstrates application of theoretical knowledge in practical tasks	-	-	-	-
PC2. Adapts to workplace requirements and follows instructions	-	-	-	-
PC3. Demonstrates clear understanding of concepts during viva	-	-	-	-
NOS Total	-	-	30	30

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1610
NOS Name	OJT/Project & Viva (60 hrs.)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Software Developer
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

HAR/ITS/N1608: Programming with Python (ACCA)

Description

The NOS introduces learners to the fundamentals and advanced concepts of Python programming. It covers Python environment setup, syntax, variables, operators, control statements, data structures, functions, file handling, modules, object-oriented programming, exception handling, regular expressions, and database connectivity with MySQL.

Scope

The scope covers the following :

- Provide learners with comprehensive knowledge and practical skills in Python programming, covering fundamentals, control structures, data handling, functions, file management, modules, object-oriented programming, exception handling, regular expressions, and database connectivity with MySQL.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Install and configure Python environment and develop programs using correct syntax, variables, operators, and data types.
- PC2.** Apply control statements, loops, functions, and data structures to solve programming problems efficiently.
- PC3.** Perform file handling, module creation, package management, and exception handling in Python applications.
- PC4.** Implement object-oriented programming concepts such as classes, inheritance, encapsulation, and method overriding.
- PC5.** Develop database-driven applications using Python and MySQL with CRUD operations and regular expression validation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamentals of Python programming including syntax, variables, data types, operators, control statements, and data structures.
- KU2.** Concepts of functions, modules, packages, object-oriented programming, exception handling, and regular expressions in Python.
- KU3.** Techniques for file handling, database connectivity, MySQL operations, and development of database-driven applications.
- KU4.** Software development practices, debugging methods, coding standards, and safe handling of program data and files.
- KU5.** Organizational requirements related to documentation, teamwork, problem solving, data security, and professional coding practices.



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read, understand, and prepare technical documents, program codes, and project reports effectively.
- GS2.** Communicate clearly with team members, trainers, and clients regarding technical requirements and solutions.
- GS3.** Apply analytical thinking and problem-solving skills for debugging and developing efficient Python applications.
- GS4.** Demonstrate planning, organizing, and time management skills while completing programming tasks and projects.
- GS5.** Follow professional ethics, teamwork, customer-centric approach, and decision-making practices in software development activities.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	22	15	10	-
PC1. Install and configure Python environment and develop programs using correct syntax, variables, operators, and data types.	-	-	-	-
PC2. Apply control statements, loops, functions, and data structures to solve programming problems efficiently.	-	-	-	-
PC3. Perform file handling, module creation, package management, and exception handling in Python applications.	-	-	-	-
PC4. Implement object-oriented programming concepts such as classes, inheritance, encapsulation, and method overriding.	-	-	-	-
PC5. Develop database-driven applications using Python and MySQL with CRUD operations and regular expression validation.	-	-	-	-
NOS Total	22	15	10	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1608
NOS Name	Programming with Python (ACCA)
Sector	Information Technology Sector
Sub-Sector	
Occupation	Software Developer
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Qualification Pack

HAR/ITS/N1609: Core Java

Description

The NOS Covers Java fundamentals, OOP concepts, packages, exception handling, file handling, collections, multithreading, AWT, applets, and Swing for developing console-based and GUI-based Java applications.

Scope

The scope covers the following :

- Develops skills in Java programming for creating object-oriented, multithreaded, file-based, and GUI applications using Core Java technologies.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Describe the features of Java programming language
- PC2.** Define and use basic syntax, semantics & structure of code in Java
- PC3.** Define various keywords, data types, operators and control flow statements in Java.
- PC4.** Use arrays and Strings in Java programming
- PC5.** Define and use the concept of encapsulation, inheritance, polymorphism and abstraction in Java programming.
- PC6.** Use exceptional handling mechanism to handle errors in Java programming.
- PC7.** Design GUI-based applications using AWT components and event handling mechanisms.
- PC8.** Create and execute applets with proper lifecycle methods and graphical components.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The organization's requirement, goal and scope of work.
- KU2.** Knowledge of Java programming concepts, syntax, and object-oriented programming principles.
- KU3.** Knowledge of exception handling, file handling, collections framework, and multithreading concepts.
- KU4.** Understanding of GUI development using AWT, Applets, and Swing components.
- KU5.** Understanding of debugging, testing, and maintaining Java applications in a professional work environment.
- KU6.** Awareness of workplace safety, data security, and quality standards in software development projects.

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Identify client needs and deliver the right solution.
- GS2.** Break down complex problems into smaller tasks.
- GS3.** Communicate effectively with team members and clients.
- GS4.** Manage time, resources, and tasks to meet deadlines.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	22	15	10	-
PC1. Describe the features of Java programming language	-	-	-	-
PC2. Define and use basic syntax, semantics & structure of code in Java	-	-	-	-
PC3. Define various keywords, data types, operators and control flow statements in java.	-	-	-	-
PC4. Use arrays and Strings in java programming	-	-	-	-
PC5. Define and use the concept of encapsulation, inheritance, polymorphism and abstraction in java programming.	-	-	-	-
PC6. Use exceptional handling mechanism to handle errors in java programming.	-	-	-	-
PC7. Design GUI-based applications using AWT components and event handling mechanisms.	-	-	-	-
PC8. Create and execute applets with proper lifecycle methods and graphical components.	-	-	-	-
NOS Total	22	15	10	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HAR/ITS/N1609
NOS Name	Core Java
Sector	Information Technology Sector
Sub-Sector	
Occupation	Software Developer
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment shall be conducted online upon completion of the course and will include MCQs, subjective and practical questions.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HAR/ITS/N0909.Basics of Information Technology	11	15	10	-	36	12
HAR/ITS/N0407.Operating System (ACCA)	9	0	5	-	14	4
HAR/ITS/N1604.Introduction to Data representation and Digital Logic	9	0	5	-	14	4
HAR/ITS/N1204.Basics of Web Page Designing (45 hrs.)	18	15	10	-	43	14
HAR/ITS/N1603.Data Communications and Networking Concepts	9	0	5	-	14	5
HAR/ITS/N1601.Object-Oriented Programming and Database Management System	22	15	10	-	47	16
HAR/ITS/N1602.Introduction to Software Development Life Cycle (SDLC)	9	0	5	-	14	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	4
HAR/ITS/N1610.OJT/Project & Viva (60 hrs.)	0	0	30	30	60	20
Total	107	75	80	30	292	84

Elective: 1 Advanced Programming: Programming with Python

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HAR/ITS/N1608.Programming with Python (ACCA)	22	15	10	-	47	16
Total	22	15	10	-	47	16

Elective: 2 Advanced Programming: Core Java



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HAR/ITS/N1609.Core Java	22	15	10	-	47	16
Total	22	15	10	-	47	16



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.